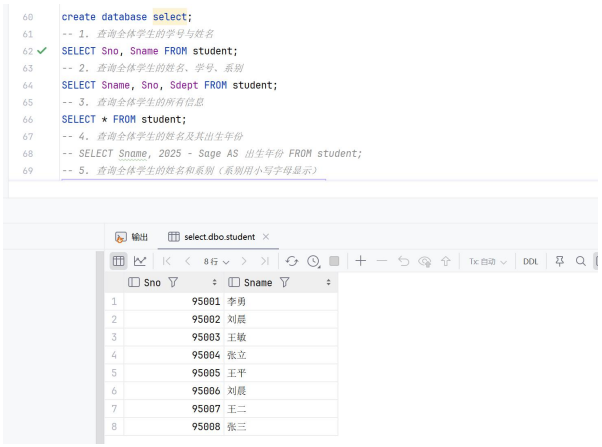
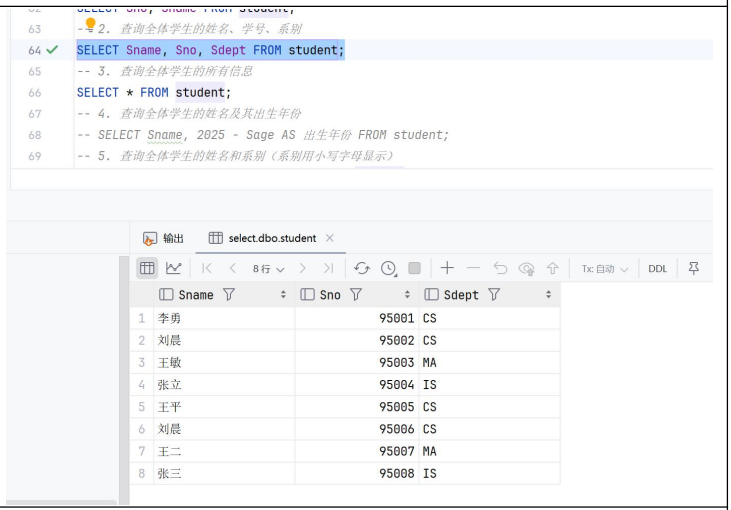
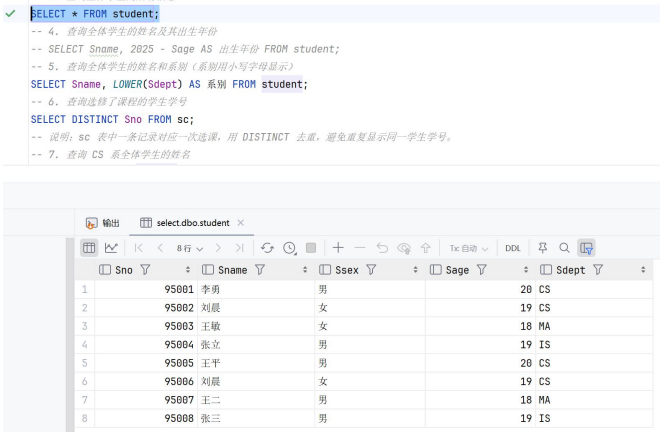
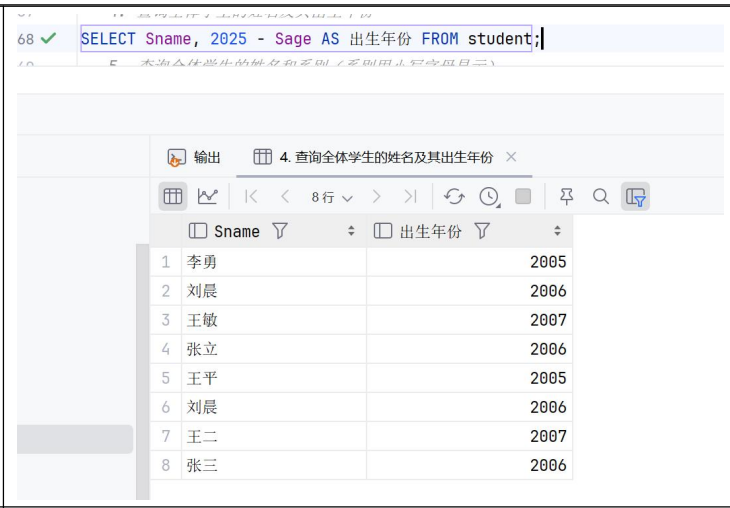
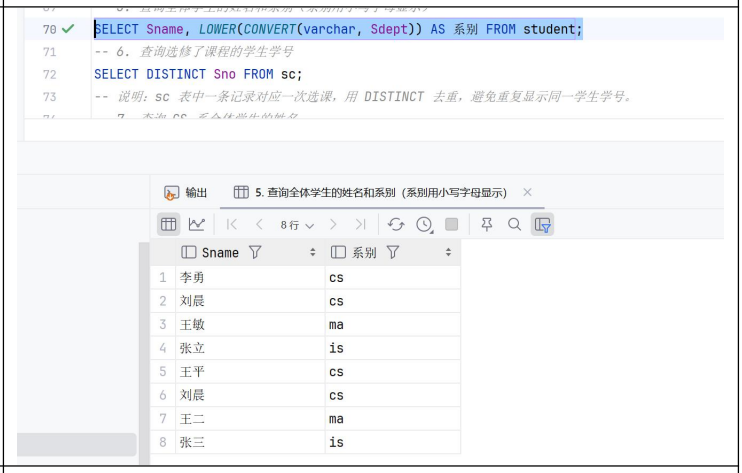
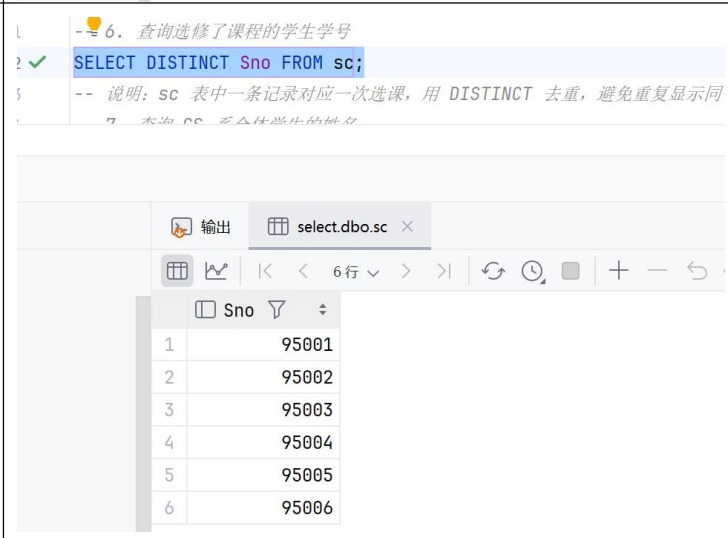
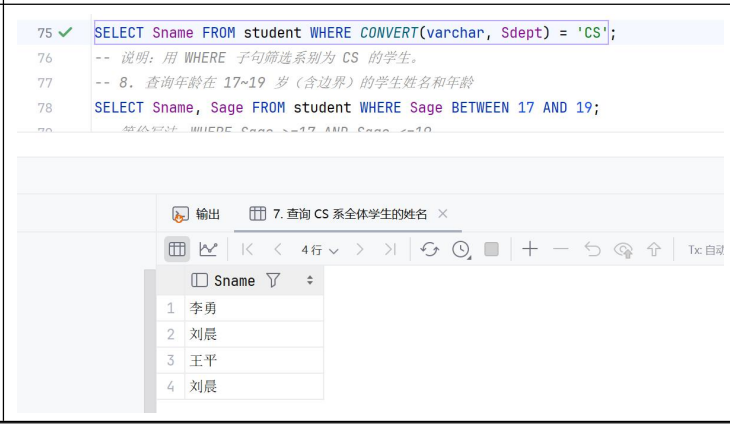
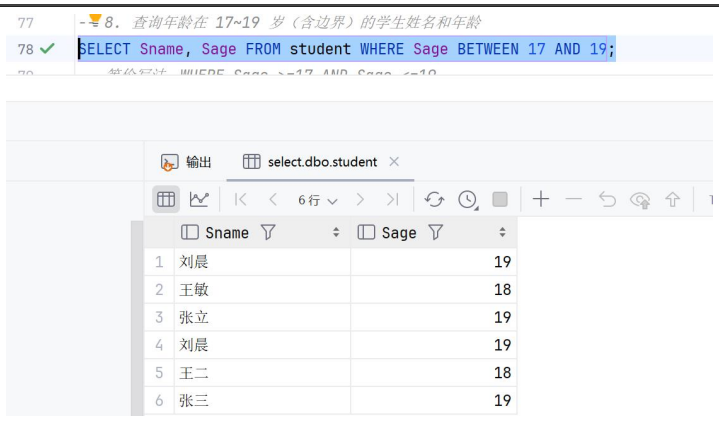
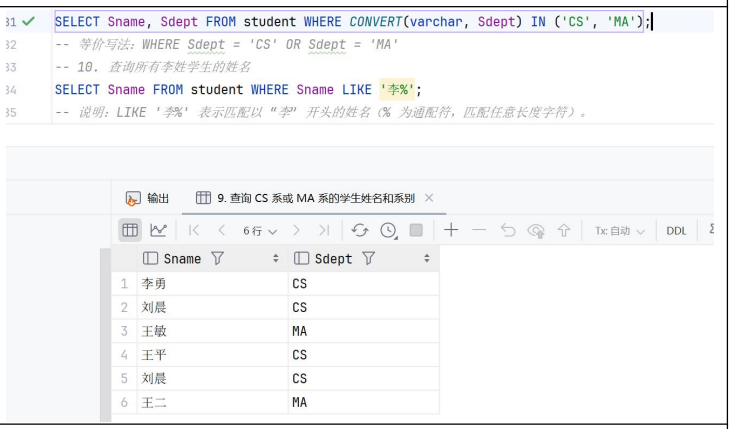
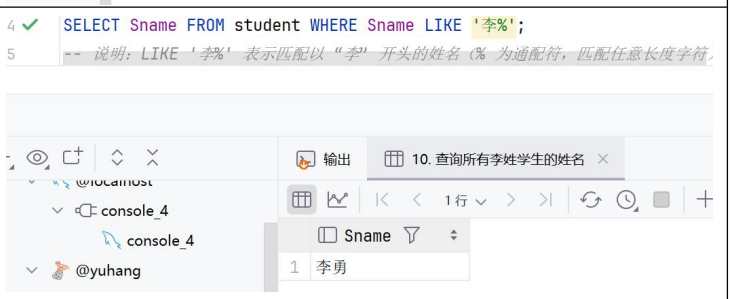
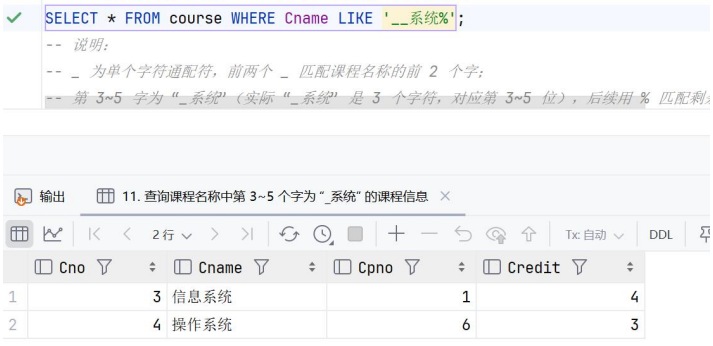
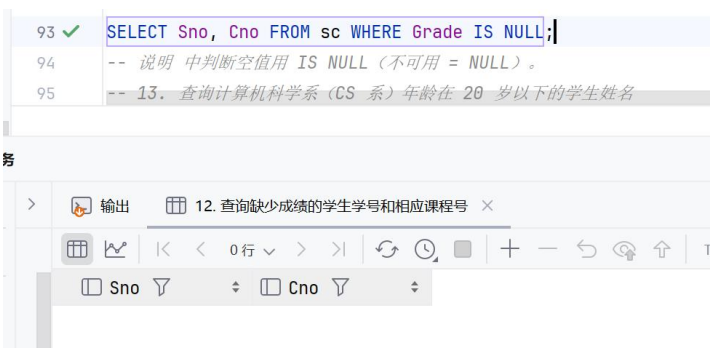
《数据库原理及应用》实验报告					
实验题目				**日期**	年 月 日
班级		**姓名**		**成绩**	
实验环境	SQL Server 2008				

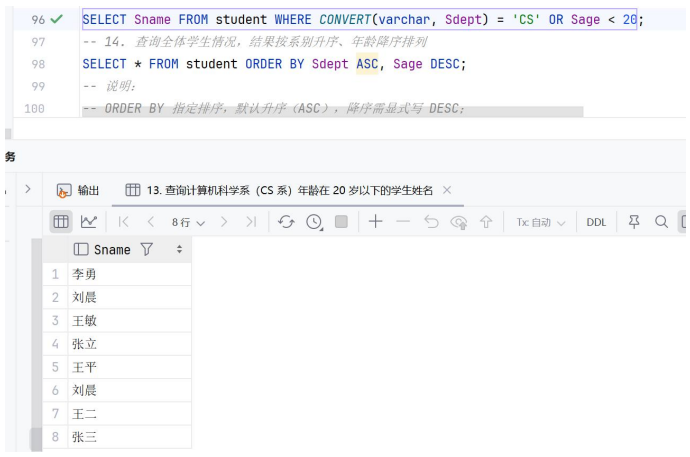
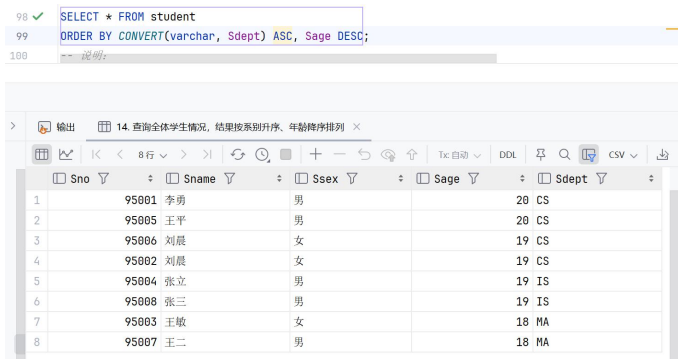
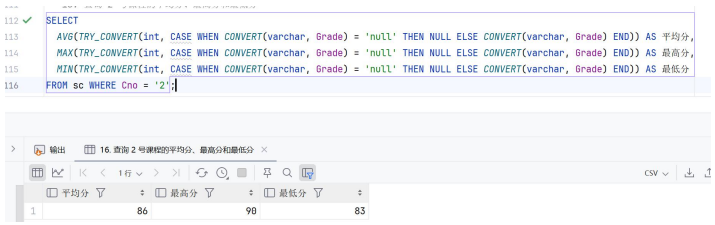
实验内容

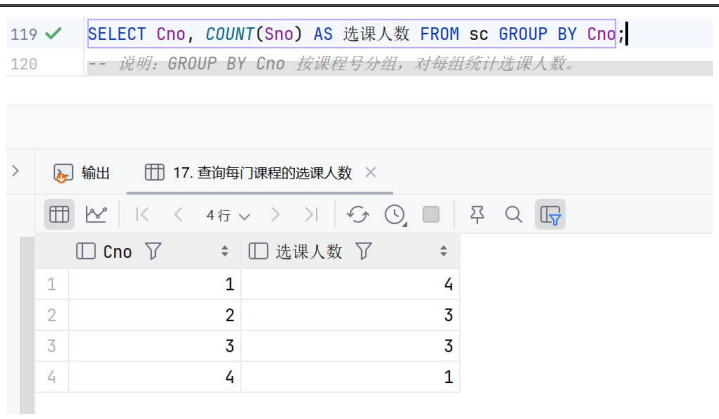
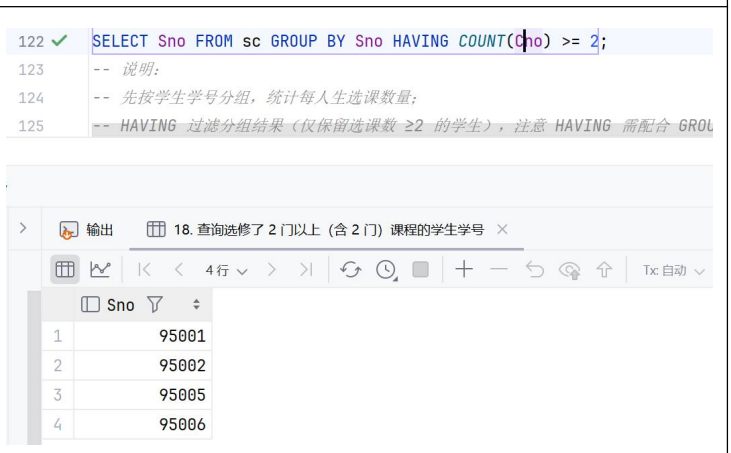
一、基础单表查询（基于 Student/Course/SC 单表）

序号	题目	对应 T-SQL 答案	操作照片
1	查询全体学生的学号与姓名	SELECT Sno, Sname FROM Student;	
2	查询全体学生的姓名、学号、系别	SELECT Sname, Sno, Sdept FROM Student;	
3	查询全体学生的所有信息	SELECT * FROM Student;	

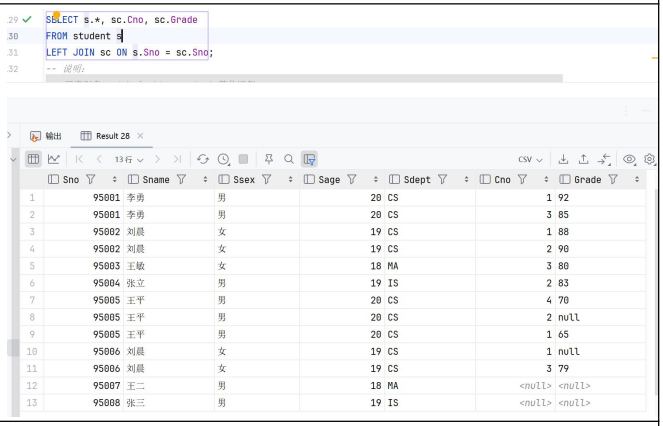
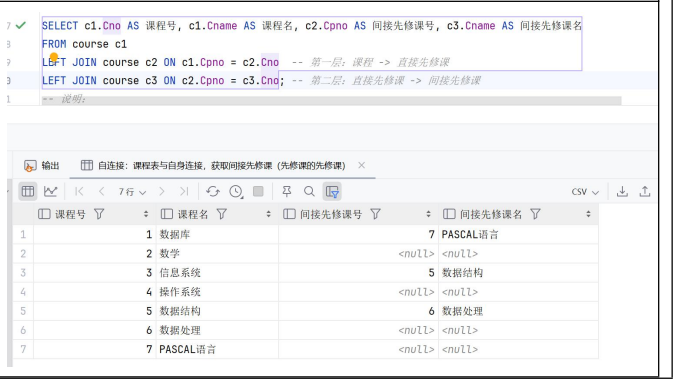
4	查询全体学生的姓名及其出生年份	SELECT Sname, 2025 - Sage AS 出生年份 FROM student;	
5	查询全体学生的姓名和系别（系别用小写字母显示）	SELECT Sname, LOWER(CONVERT(varchar, Sdept)) AS 系别 FROM student;	
6	查询选修了课程的学生学号（去重）	SELECT DISTINCT Sno FROM SC;	
7	查询 CS 系全体学生的姓名	SELECT Sname FROM student WHERE CONVERT(varchar, Sdept) = 'CS';	

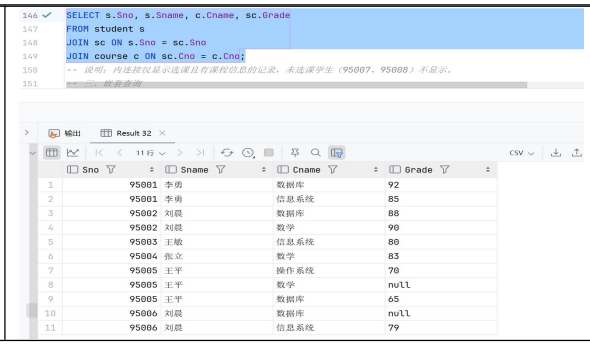
8	查询年龄在 17～19 岁（含边界）的学生姓名和年龄	SELECT Sname, Sage FROM student WHERE Sage BETWEEN 17 AND 19;	
9	查询 CS 或 MA 系的学生姓名和系别	SELECT Sname, Sdept FROM student WHERE CONVERT(varchar, Sdept) IN ('CS', 'MA');	
10	查询所有李姓学生的姓名	SELECT Sname FROM Student WHERE Sname LIKE '李%';	
11	查询课程名称中第 3～5 个字为“系统”的课程信息（前 2 字任意）	SELECT * FROM Course WHERE Cname LIKE '__系统%';	
12	查询缺少成绩的学生学号和相应课程号	SELECT Sno, Cno FROM SC WHERE Grade IS NULL;	

13	查询计算机科学系（CS）年龄在 20 岁以下的学生姓名	SELECT Sname FROM student WHERE CONVERT(varchar, Sdept) = 'CS' OR Sage < 20;	
14	查询全体学生情况，结果按系别升序、年龄降序排列	SELECT * FROM student ORDER BY CONVERT(varchar, Sdept) ASC, Sage DESC;	
15	查询选课记录总数及选修了课程的学生人数	SELECT COUNT(*) AS 选课记录总数, COUNT(DISTINCT Sno) AS 选课学生人数 FROM SC;	
16	查询 2 号课程的平均分、最高分和最低分	SELECT AVG(TRY_CONVERT(int, CASE WHEN CONVERT(varchar, Grade) = 'null' THEN NULL ELSE CONVERT(varchar, Grade) END)) AS 平均分, MAX(TRY_CONVERT(int, CASE WHEN CONVERT(varchar, Grade) = 'null' THEN NULL ELSE CONVERT(varchar, Grade) END)) AS 最高分, MIN(TRY_CONVERT(int, CASE WHEN CONVERT(varchar, Grade) = 'null' THEN NULL ELSE CONVERT(varchar, Grade) END)) AS 最低分 FROM sc WHERE Cno = '2';	

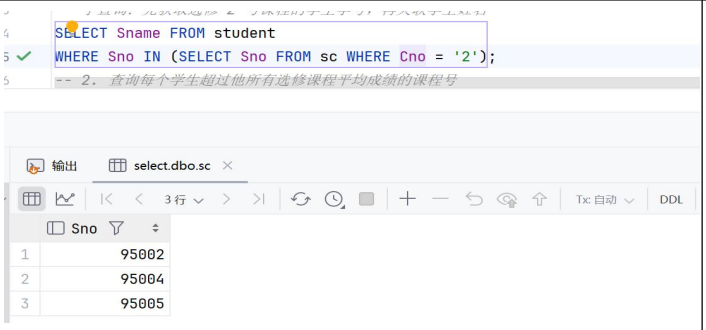
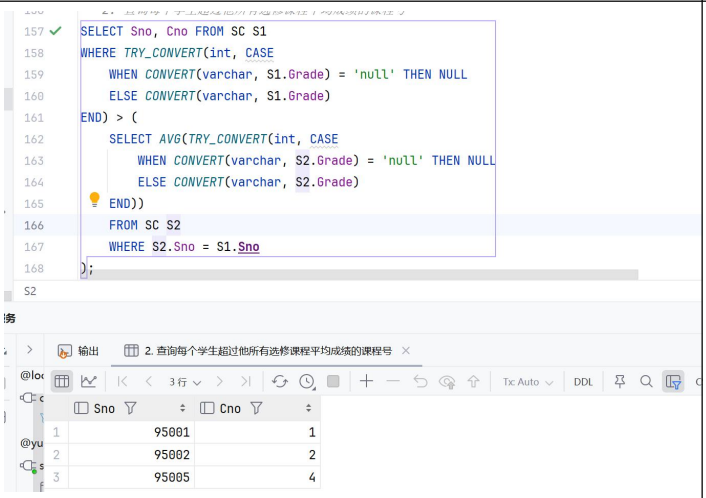
17	查询每门课程的选课人数	SELECT Cno, COUNT(Sno) AS 选课人数 FROM SC GROUP BY Cno;	
18	查询选修了 2 门以上课程的学生学号	SELECT Sno FROM SC GROUP BY Sno HAVING COUNT(Cno) > 2;	

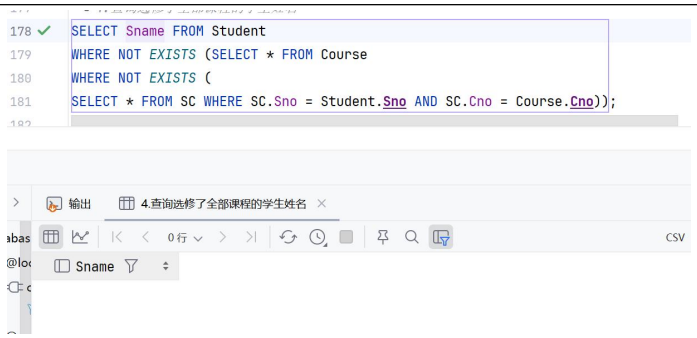
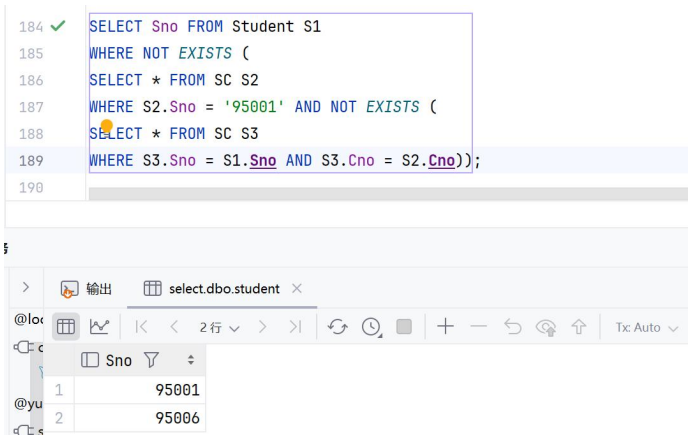
二、连接查询（多表关联）

序号	题目	对应 T-SQL 答案	操作照片
1	查询每个学生及其选修课程的情况（含未选课学生）	SELECT Student.*, SC.* FROM Student LEFT JOIN SC ON Student.Sno = SC.Sno;	
2	查询每门课的间接先修课（先修课的先修课）	SELECT C1.Cno, C1.Cname, C2.Cpno AS 间接先修课 FROM Course C1 JOIN Course C2 ON C1.Cpno = C2.Cno;	

3	查询每个学生的学号、姓名、选修课程名及成绩	SELECT Student.Sno, Student.Sname, Course.Cname, SC.Grade FROM Student JOIN SC ON Student.Sno = SC.Sno JOIN Course ON SC.Cno = Course.Cno;	

三、嵌套查询（子查询）

序号	题目	对应 T-SQL 答案	操作照片
1	查询所有选修了 2 号课程的学生姓名	SELECT Sname FROM Student WHERE Sno IN (SELECT Sno FROM SC WHERE Cno = '2');	
2	查询每个学生超过他所有选修课程平均成绩的课程号	SELECT Sno, Cno FROM SC S1 WHERE TRY_CONVERT(int, CASE WHEN CONVERT(varchar, S1.Grade) = 'null' THEN NULL ELSE CONVERT(varchar, S1.Grade) END) > (SELECT AVG(TRY_CONVERT(int, CASE WHEN CONVERT(varchar, S2.Grade) = 'null' THEN NULL ELSE CONVERT(varchar, S2.Grade) END)) FROM SC S2 WHERE S2.Sno = S1.Sno);	

3	查询其他系中比 CS 系所有学生年龄都小的学生姓名	SELECT Sname FROM student WHERE CONVERT(varchar, Sdept) != 'CS' -- 转换 Sdept 类型后判断非 CS 系 AND Sage < ALL (SELECT Sage FROM student WHERE CONVERT(varchar, Sdept) = 'CS');	
4	查询选修了全部课程的学生姓名	SELECT Sname FROM Student WHERE NOT EXISTS (SELECT * FROM Course WHERE NOT EXISTS (SELECT * FROM SC WHERE SC.Sno = Student.Sno AND SC.Cno = Course.Cno));	
5	查询至少选修了学号为 '95001' 学生选修的全部课程的学生学号	SELECT Sno FROM Student S1 WHERE NOT EXISTS (SELECT * FROM SC S2 WHERE S2.Sno = '95001' AND NOT EXISTS (SELECT * FROM SC S3 WHERE S3.Sno = S1.Sno AND S3.Cno = S2.Cno));	

四、集合查询

序号	题目	对应 T-SQL 答案	操作照片

1	查询CS系的学生或年龄不大于19岁的学生（合并结果，去重）	-- 筛选 CS 系学生：转换所有 text 字段 SELECT CONVERT(varchar(50), Sno) AS Sno, -- 若 Sno 是 text 类型，需转换 CONVERT(varchar(50), Sname) AS Sname, -- 若 Sname 是 text 类型，需转换 CONVERT(varchar(50), Sdept) AS Sdept, Sage FROM student WHERE CONVERT(varchar(50), Sdept) = 'CS' UNION -- 合并结果并去重 -- 筛选年龄≤19 岁的学生：保持字段类型与上半部分一致 SELECT CONVERT(varchar(50), Sno) AS Sno, CONVERT(varchar(50), Sname) AS Sname, CONVERT(varchar(50), Sdept) AS Sdept, Sage FROM student WHERE Sage <= 19;	